Zoo201 mid 2022 subject

Composed by Muhammad Habib

Answered by Qudratulah ktk

Q 1:

Platyhelminthes ki 3 classes Kon c hen???

The Platyhelminthes includes three classes: 1.the Turbellaria (free-living flatworms), 2.the Trematoda (flukes), 3.the Cestoda (tapeworms).

Q1 What is coccidiosis?

Ans:-

Coccidiosis is an intestinal tract infection caused by a single-celled organism (a protozoa) called coccidia. Coccidia are sub-classified into a number of genera (groups), and each genus (group) has a number of species.

Q 2. What is zonites?

Ans:-

The body surface of kinorhyncha is devoid of cilia and is composed of 13 or 14 definite unit which is called zonites.

Q 3. What is the benefits of cuttlebone?

Ans:-

Cuttlebone is an important dietary supplement for birds because it is a great source of necessary minerals and calcium, which helps birds with bone formation and blood clotting. Cuttlebone is an inexpensive source of calcium carbonate and other trace minerals for your bird.

Q 4. What are the similarities of Annelids and arthropods?

Ans:-

Both arthropods and annelids are segmented, and members of the annelid class Polychaeta have a pair of appendages on each segment. The plan of the nervous system in arthropods is very similar to that of annelids, and the basic plan in both groups shows a tubular, dorsal heart, which is then lost or modified in some

Q 5. Write the advantages of coelom?

Ans:-

The advantages of the coelom are as follows:

1.The organs like that of the digestive tract need more space to grow. …

2.Some organs like gonads need more space only during the breeding season.

3 Coelom also allows the formation of well-organized circulatory system with an efficient heart to draw blood from vessels.

Q. 1 kingdom animalia?

Ans:-

Definition:-

The Kingdom Animalia is a large group that consists of eukaryotic, multicellular organisms that are heterotrophic in nature.

=>Kingdom Animalia divided into two major groups:

- **Invertebrates**
- **Vertebrates**

=>Characteristics of kingdom Animalia are:

- **They are multicellular organisms which do not possess chlorophyll.**
- **They are eukaryotic organisms.**
- **Cell wall is absent.**
- **Mode of nutrition is heterotrophic**

Q 2. Characteristics of nematoda?

Ans:-

Characteristics of Nematoda:

- **Their body is bilaterally symmetrical and triploblastic.**
- **They are cylindrical in shape.**

- **They exhibit tissue level organization.**
- **Their body has a cavity or pseudocoelom.**
- **The alimentary canal is distinct, with the mouth and the anus.**
- **They are sexually dimorphic.**

Q. 3 Oligochaetes?

Ans:-

Oligochaeta is a subclass of animals in the phylum Annelida, which is made up of many types of aquatic and terrestrial worms, including all of the various earthworms.

Q1. What is Cnidocytes?

Ans:-

A cnidocyte is an explosive cell containing one giant secretory organelle called a cnidocyst that can deliver a sting to other organisms. The presence of this cell defines the phylum Cnidaria.

Q 2. Earthworm?

Ans:-

An earthworm is a terrestrial invertebrate that belongs to the phylum Annelida. They exhibit a tube-within-a-tube body plan, are externally segmented with corresponding internal segmentation, and usually have setae on all segments. … Earthworms are commonly found in soil, eating a wide variety of organic matter.

Q 3. Three major clades of deutrostomes:-

Ans:-

The three major clades of deuterostomes are the echinoderms, the hemichordates, and the chordates.

Q 4. Life cycle of Apicomplexans?

Ans:-

The apicomplexa have complex life cycles that are characterized by three distinct processes: sporogony, merogony and gametogony.

Q 5. Evolutionary adaptations of parasitic nematodes?

Ans:-

The evolutionary success and ubiquitous dis- tribution of parasitic nematodes is associated with elaborate physical and biochemical adaptations to parasitism: from a tough collagenous outer cuticle and sophisticated attachment and invasion appa- ratus such as hooks and suckers, to the expression of immunomodulatory.

Prepard by Qudrat ktk

Q 1 characteristics of Annelids?

Ans:-

Characteristics of Annelida

1.The Annelids are coelomate and triploblastic.

2.They exhibit organ system level organization.

3.Their body is segmented.

4.They respire through their body surface.

Q 2. Define siphon?

Ans:-

Definition:-

The tubular organ through which water is ejected from the gill cavity of a cephaloid.

Q 3. Circulatory system of cephalopod?

Ans:-

The core of the cephalopod circulatory system is a series of three beating hearts. This trio of hearts connects to a high pressure system of veins, arteries, and capillaries – unique among all mollusks. These are almost like the parts of the human heart that pump blood to our lungs.

Q 4. Monoecious and dioecious?

Ans:-

Monoecious:-

Monoecious organisms are those containing both male and female reproductive organs.

Dioecious:-

These are organisms that produce either male or female reproductive organs and gametes but never both at the same time.

Q 1. Define pseudocoelomates with example?

Answer:-

Definition:-

A pseudocoelomate is an organism with body cavity that is not derived from the mesoderm, as in a true coelom, or body cavity.

Examples:-

The nematodes, rotifers, gastrotrichs, and introverts.

Q 2. Life cycle of Apicomplexans?

Ans:-

Most Apicomplexans have a complex life-cycle, involving both asexual and sexual reproduction. Typically, a host is infected by ingesting cysts, which divide to produce sporozoites that enter the host's cells. Eventually, the cells burst, releasing merozoites which infect new host cells.

Q 3. Two common characteristics of Annelids and arthropods?

Ans:-

1.Both Annelida and Arthropoda consist of segmented body structures.

2.Both Annelida and Arthropoda consist of a hydrostatic skeleton.

Q 4. Copulatory function in oligochaeta?

Ans:-

Sexually mature oligochaetes have a clitellum, which is a modification of a section of the body wall consisting of a glandular, saddlelike thickening near the gonopores. During copulation, the clitellum secretes a mucus that keeps the worms paired while sperm are being exchanged.

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Q 1. Adaptation of practice nematodes?

Ans;-

The integument of the parasite have lost epidermis and have developed several layered thick protective covering of cuticle. The cuticle is resistent to host's digestive enzymes, antitoxin and abrasive action of the food and roughage passtng through the digestive tract.

Q 2. Monophyletic group and polyphyletic group?

Ans:-

Monophyletic group:-

A monophyletic taxon is one that includes a group of organisms descended from a single ancestor.

Polyphyletic group:- Apolyphyletic taxon is composed of unrelated organisms descended from more than one ancestor.

Q 3. Metamerism:-

Ans:-

Metamerism is the repetition of homologous body segments. This type of development can be seen in the Annelids, which include earthworms, leeches, tubeworms, and their relatives. It is also seen in a more advanced form in the Arthropods, such as crustaceans, insects, and their relatives.

Q 4. Protostome?

Ans:-

The term Protostomia (from the Greek "proto," meaning first, and "stoma," meaning mouth) was coined by the biologist Karl Grobben in 1908. It distinguishes a group of invertebrate animals based upon the fate of the blastopore (the first opening of the early digestive tract) during embryonic development.

Q 5. Characteristics of kingdom animalla?

Ans:-

1.These organisms are multicellular, eukaryotic and without chlorophyll.

2.The cells possess no cell walls and plastids.

3.Central vacuoles are absent but small vacuoles may occur.

4.Most of them are free moving (except sponges and some coelentrates)

Nutrition is primarily ingestive.

5.Reproduction is generally sexual and the haploid stage is represented only by gametes.

6 Growth of organisms stops when the adult stage is reached

***Arthropods/Annelids difference* 5marks**

Arthropods are invertebrate animals that possess a segmented body, an exoskeleton, and

Paired appendages. Whereas, annelids are another group of invertebrates that are segmented

Worms by transverse rings. Though arthropods have an exoskeleton, annelids do not.

Furthermore, arthropods have an open circulatory system while annelids have a closed

Circulatory system. Moreover, arthropods have a chitinous cuticle while annelids have a

Collagenous cuticle. Insects, crustaceans, arachnids are several groups of arthropods while

Earthworms, ragworms, and leeches are some groups of annelids. Thus, this summarizes the Difference between arthropods and annelids.

***Body wall of turbellarian*5marks**

Body wall:

A basement membrane that separates the epidermis from mesodermally derived tissues. An Outer layer of circular muscles and inner layer of longitudinal muscle lie beneath the basement Membrane. The innermost layer is endodermis that comprises the digestive cavity. Rhabdites A protective mucus sheath around the body possibly in response to attempted predation or Desiccation. Adhesive glands attract part of turbellaria to the substrate. Releaser glands release The enzyme that dissolved the attachment.

Locomotion:

They move using cilia using muscular undulation.

Digestion and Nutrition:

Most turbellarians are carnivores, some are herbivores. Food Digestion is extracellular and intracellular.

***monoecious/diecious species* 3 marks**

Dioecious species are either male or female. Each individual, when it produces reproductive Structures, produces only male or female structures.

***Function of pellicle in protozoa* 2 marks**

Most protozoans are enclosed by a skeletal structure known as the pellicle consisting of the

Plasma membrane and underlying cytoskeleton (with additional membranes, microtubules,

Microfilaments, or plates of cellulose or protein). The pellicle maintains the shape of the cell.

Define typhlosole 2 marks

A dorsal fold of the luminal epithelium called the typhlosole substantially increases the surface

Area of the

Intestine.

Tubular system 2marks

Parasitic nematodes have a more advances system, called the tubular system that developed

From the renette system. In this system, the renettes unite to form a large canal which opens

Outside through an excretory pore.

Long Q marks

Characteristics of phylum

Cnidaria include:

1. **Radial or biradial symmetry**
2. **Diploblastic, tissue-level organization**
3. **Gastrovascular cavity**
4. **Gelatinous mesoglea between the epidermal and gastrodermal tissue**

Layers.

5. **Nervous system in the form of a nerve net.**
6. **Specialized cells called cnidocytes, used in defense, feeding, and attachment.**
7. **Endocrine have hormones, secreted by**

Nervous system.

8. **No circulatory system, use cellular diffusion and mesoglea.**
9. **Ciliated planula larva is found.**
10. **Members of the phylum Cnidarian (Gr. Knide means nettle) possess radial or biradial**

Symmetry.

Characteristics of deuterostomes:

Deuterostomes are also known as

Enterocoelomates because their coelom

Develops through enterocoely.

There are three major clades of deuterostomes:

1. **Chordata (vertebrates and their kin)**
2. **Echinodermata (starfish, sea urchins,**

Sea, cucumbers, etc.)

• 3. Hemichordata (acorn warms and graptolites). Extinct deuterostome groups may include the

Phylum Vetulicolia.

It is a sister clade of Protostomia, with

Which forms the Nephrozoa clade.

Parasitic Nematoda show evolutionary adoption?

Parasitic nematodes show a number of

Evolutionary adaptations to the way of their life. These adaptations are:

• 1. They have high reproductive potential.

• 2. Their life cycles increase the likelihood of transmission from one host to another.

• 3. They develop an enzyme resistant cuticle, resistant eggs, and encysted larvae.

Q No2

Difference between monophyletic, Diphyletic and polyphyletic?

• A fundamental question concerning animal origins is whether animals are;

• Monophyletic (derived from a single ancestor)

• Diphyletic (derived from two ancestors)

• Polyphyletic (derived from many ancestors).

Monophyletic:

The goal of animal systematics is to arrange animals into groups that reflect evolutionary Relationships.

• Ideally, these groups should include a single ancestral species and all of its descendants.

• Such a group is called a monophyletic group.

Diphyletic:

Some zoologists believe the animal kingdom to be at least diphyletic, with the porifera being Derived separately from all other phyla.

• The similarity of poriferan choanocytes and choanoflagellates protists suggest Evolutionary ties between these groups.

Polyphyletic:

These groups have members that can be traced to separate ancestors.

What is spongocoel?

Ostia are the outer openings of porocytes and lead directly to a chamber called the spongocoel.

It is a single, large, spacious central cavity in the sponge's body.

Define Determinate cleavage?

Determinate cleavage:

A second characteristic common to many protostomes is that early cleavage is determinate.

• Determinate means that the fate the cells is established very early in development.

Q No

Circulatory system in cephalopod:

Cephalopods have a closed circulatory

System.

• Blood is confined to vessels throughout its circuit around the body.

• Capillary beds connect arteries and veins, and exchanges of gases, nutrients, and metabolic

Wastes occur across capillary walls.

• In addition to having a heart consisting of two auricles and one ventricle, cephalopods have

Contractile arteries

And structures called branchial hearts.

The latter are at the base of each gill and help move blood through the gill.These modifications

Increase blood pressure and the rate of blood flow—necessary for active animals with relatively

High metabolic rates.

• Large quantities of water circulate over the gills at all times.

• Cephalopods exhibit greater excretory efficiency because of the closed circulatory system.

Q no.

Role of osphradia:

Osphradia are chemoreceptors in the anterior wall of the mantle cavity that detect sediment and

Chemicals in inhalant water or air.

• The osphradia of predatory gastropods help detect prey.

QNo.

Definition of siphon:

Some gastropods have a rolled extension of the mantle, called a siphon, that serves as an

Inhalant tube.

Q no. Phylum Nematoda

Phylum Nematoda is a phylum of invertebrate animals. They are comprised of roundworms (oo

Nematodes). They are identifiable by their smooth and tough outer covering and narrow

Cylindrical unsegmented body. Their name is etymologically derived from the Greek word nema,

Nematos, meaning "thread.

QNo.

Ecdysis is the process of an arthropod moulting its exoskeleton. Moulting is a critical but

Vulnerable time for arthropods. Their existing cuticle weakens at specific points and by taking in

Water or air the animal splits its old exoskeleton. The animal then extracts itself from the old skin

And inflates its new skin.

Summary of Molting

Step 1: Apolysis: separation of old exoskeleton from epidermis

Step 3: Production of cuticulin layer for new exoskeleton

Step 4: Activation of molting fluid

Step 5: Digestion and absorption of old endocuticle.

Define Ecdysis and it's four stages?

The growth of an arthropod would be

Virtually impossible unless the exoskeleton

Were periodically shed, such as in the molting process called ecdysis

(Gr. Ekdysis, getting out).

Four stages:

Ecdysis is divided into four stages:

• (1) Enzymes, secreted from hypodermal glands, begin digesting the old procuticle to separate

The hypodermis and the exoskeleton.

(2) new procuticle and epicuticle are secreted.

• (3) the old exoskeleton splits open along predetermined ecdysial lines when the animal

Stretches by air or water intake; pores in the procuticle secrete additional epicuticles.

(4) Finally, calcium carbonate deposits and/or sclerotization harden the new exoskeleton.

Q No.

Pinacocytes

The process in which food is engulf in form of liquid.

Q No.

Schizocoel hypothesis about the Origin of coelom:

Schizocoel hypothesis: Coelom is formed by the splitting of mesoderm i.e.

Protostomes. It suggests that a triploblastic acoelomate (flatworm) body form is the

Ancestor of the coelomate.

QNo.

Heartworm ?

Heartworm disease is a serious disease that results in severe lung disease, heart failure, other

Organ damage, and death in pets, mainly dogs, cats, and ferrets. It is caused by a parasitic

Worm called Dirofilaria immitis. The worms are spread through the bite of a mosquito.

QNo.

Mastax?

Pharynx contains a structure called the mastax (jaws). The mastax is a muscular organ. It ggrind

Food. The inner walls of the mastax contain several sets of jaws called trophi. The trophi have

Different structures.

Q no

Hydra reproduction

The common asexual method of reproduction by hydras is budding. Buds originate at the

Junction of the stalk and gastric aregions. The bud begins as a hemispherical outpouching that

Eventually elongates, becomes cylindrical, and develops tentacles. The bud then pinches off

And a new individual becomes independen..